Alexander D. Kaiser

Computational cardiovascular engineering, Stanford University
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Education

New York University, Courant Institute of Mathematical Sciences
PhD Mathematics. Advisor: Charles Peskin. Thesis: Modeling the Mitral Valve 2017

MS Mathematics. Advisor: Jonathan Goodman. 2013
Thesis: Computational Experiments in Markov Chain Monte Carlo

University of California, Berkeley
BA Mathematics, minor in Computer Science 2009

Grants

- **K25 Mentored Quantitative Research Development Award, $908,880** 2024 – 2029
  National Heart, Lung, and Blood Institute (NHLBI), National Institutes of Health (NIH)
- **American Heart Association Career Development Award, $231,000** 2024 – 2027
- **Instructor K Award Support, $100,000** 2025 – 2027
  Stanford Maternal & Child Health Research Institute

Awards and Fellowships

- Kurt O. Friedrichs Prize for Outstanding Dissertation in Mathematics 2018
  Awarded annually to at most two PhD graduates, Department of Mathematics, New York University
- Math Master's Thesis Prize 2014
  Awarded annually to at most two master's graduates, Department of Mathematics, New York University
- Mechanisms and Innovation in Cardiovascular Disease T32 Training Fellowship, NHLBI, NIH 2018
- National Science Foundation (NSF) Graduate Research Fellowship 2013
- Stanford-Arizona-Morehouse-UAB Cardiovascular Research Symposium Poster Award 2024
- Benchmark Capital Fellowship in Congenital Cardiovascular Bioengineering 2020
- Thomas Tyler Bringley Fellowship, Department of Mathematics, New York University 2016

Experience

Stanford University
Instructor (full-time research role), Cardiothoracic Surgery 2024 –
Research Engineer, Pediatrics (Cardiology) 2022 – 2024
Postdoctoral Scholar, Pediatrics (Cardiology), 2017 – 2022
Institute for Computational & Mathematical Engineering (ICME)
Mentors: Alison Marsden (engineering); Michael Ma (clinical)

- Founded and lead a research team modeling heart valves, congenital heart disease, and surgical repair. Secured $1.24M in competitive funding as principal investigator from NIH, the American Heart Association, and Stanford.
- Translated simulations to the clinic. The first five pediatric patients treated with the simulation-guided repair showed excellent operative outcomes and more consistent postoperative valve gradients than historical controls.
- Supervise two direct reports: a postdoctoral scholar funded on my grants and a medical student funded jointly by my grants and her fellowship. Co-mentor two other trainees and collaborate with five surgical residents.
- Extended elasticity-based design to bicuspid aortic and quadricuspid truncal valves for physiology studies.
- Developed simulation-guided design tools for surgical planning of valve repair using fluid-structure interaction.
- Compared full 3D, time-resolved simulated velocity fields against in vitro 4D flow MRI data, the first reported heart valve study to compute relative error over the entire flow field.
- Showed with simulations that, contrary to conventional wisdom, valve performance improves after neonatal truncus arteriosus repair without direct leaflet intervention, a trend confirmed on clinical imaging.

New York University, Courant Institute of Mathematical Sciences
Graduate Student Researcher, Department of Mathematics 2011 – 2017

- Created elasticity-based design, a nearly first-principles method that derives mitral valve geometry, material properties, and fiber orientations from mechanical equilibrium with only limited experimental inputs, such as loading pressure and expected stretch.
- Performed immersed boundary fluid-structure interaction simulations of the mitral valve that opened freely and closed competently across multiple cardiac cycles.
- Parallelized affine-invariant MCMC samplers on GPUs for parameter estimation in differential equations.

Innovein Medical
Consultant, Medical Device Design 2018 – 2019

- Used fluid-structure interaction simulations to design a new feature for a prosthetic vein valve, removing a major failure mode seen in prototypes. The feature was adopted in subsequent device designs, and the device later entered a clinical feasibility trial.

Lawrence Berkeley National Laboratory
Computer Systems Engineer, Complex Systems & Future Technologies Groups 2009 – 2011

- Led development of a testbed of computational reference kernels for parallel computing research.
- Maintained arbitrary- and extended-precision arithmetic libraries ARPREC and QD, including user support.

University of Newcastle, Australia
Visiting Researcher, Centre for Computer-Assisted Research Mathematics & its Applications 2011

- Designed and implemented algorithms for automated symbolic simplification of constants of the form $\sum_{i=1}^{n} \alpha_i z_i$ with α_i rational, z_i complex, and n large, released open-source as SimplifySum.
- Achieved over 1000x speedup compared to built-in algorithms in *Mathematica* on certain problems.

Apple Inc.
Cooperative Researcher, Advanced Computation Group 2010

- Developed a mixed-language interface for MatrixFFT, Apple's high-performance Fast Fourier Transform library.

Jet Propulsion Laboratory, NASA
Summer Student Researcher, Information Processing Group 2007

- Designed, implemented, and validated schemes that greatly reduced undetected errors (corrupted data passing as correct) in error-correcting codes for the NASA Deep Space Network standard.

Media Coverage

- "Baby on Hospice Receives Second Chance With Lifesaving Heart Surgery," 2025
  Healthier, Happy Lives Blog, Stanford Medicine Children's Health. article

Publications

† denotes trainee mentored by ADK

Selected highlight publications:

19. Bornemann KM†, Choi PS, Huber JA, Reed AK, Sharir A, Maskatia SA, Marsden AL, Ma MR, **Kaiser AD**. Simulations Predict Improved Valve Performance Without Direct Leaflet Intervention After Neonatal Truncus Arteriosus Repair. *The Journal of Thoracic and Cardiovascular Surgery*, 2026. arXiv doi movies
18. **Kaiser AD**, Haidar MA, Choi PS, Sharir A, Marsden AL, Ma MR. Simulation-Based Design of Bicuspidization of the Aortic Valve. *The Journal of Thoracic and Cardiovascular Surgery*, 2024. arXiv doi movies
17. **Kaiser AD**, McQueen DM, Peskin CS. Modeling the Mitral Valve. *International Journal of Numerical Methods in Biomedical Engineering*, 2019. arXiv doi movies
16. Choi PS, **Kaiser AD**, Nilkant R, Sharir A, Siddiqi U, McMullen HL, Marsden AL, Ma MR. Simulation-Guided Outlet Valve Repair: Early Integration of Computationally-Derived Principles of Free-Edge Length in Bicuspidization Repair. *Frontiers in Cardiovascular Medicine*, 2026. doi

Additional journal publications:

15. Martinez ES†, **Kaiser AD**, Reed AK, Stocker SW, Sharir A, Choi PS, Maskatia SA, Ma MR, Marsden AL. Semi-Automated Generation and Hemodynamic Assessment of Surgical Baffle Geometry for Biventricular Repair. *Annals of Biomedical Engineering. In press.* 2026. arXiv
14. Munshi SK, Huber JA, Sharir A, Ono Y, Choi PS, Reed AK, **Kaiser AD**, Ma MR. A Novel Ex-Vivo Model of Unbalanced Common Atrioventricular Valve: Insights into Biomechanics and Hemodynamics During Single Ventricle Palliation. *The Journal of Thoracic and Cardiovascular Surgery. Online ahead of print.* 2026. doi
13. **Kaiser AD**, Choi PS, Sharir A, Marsden AL, Ma MR. Simulation-Guided Design of Leaflet Height in Bicuspidization of the Aortic Valve. *JTCVS Open*, 2025. doi
12. **Kaiser AD**, Wang J, Brown AL, Zhu E, Hsiai TK, Marsden AL. A Fluid-Structure Interaction Model of the Zebrafish Aortic Valve. *Journal of Biomechanics*, 2025. arXiv doi
11. Choi PS, Sharir A, Ono Y, Shibata M, **Kaiser AD**, Palagani Y, Marsden AL, Ma MR. Combined Simulation and Ex-vivo Assessment of Free-edge Length in Bicuspidization Repair for Congenital Aortic Valve Disease, *JTCVS Open*, 2024. doi
10. Choi PS, Sharir A, Ono Y, Shibata M, **Kaiser AD**, Zhu Y, Marsden AL, Woo YJ, Ma MR, Kim JB. Effect of Graft Sizing in Valve-sparing Aortic Root Replacement for Bicuspid Aortic Valve: The Goldilocks Ratio. *The Journal of Thoracic and Cardiovascular Surgery Techniques*, 2024. doi
9. **Kaiser AD***, Schiavone NK*, Elkins CJ, McElhinney DB, Eaton JK, Marsden AL. Comparison of Immersed Boundary Simulations of Heart Valve Hemodynamics against In Vitro 4D Flow MRI Data. (*contributed equally) *Annals of Biomedical Engineering*, 2023. arXiv doi
8. **Kaiser AD**, Shad R, Schiavone N, Hiesinger W, Marsden AL. Controlled Comparison of Simulated Hemodynamics across Tricuspid and Bicuspid Aortic Valves. *Annals of Biomedical Engineering*, 2022. arXiv doi movies
7. Frishman S, Kight A, Pirozzi I, Maddineni S, Imbrie-Moore A, Karachiwalla Z, Paulsen MJ, **Kaiser AD**, Woo YJ, Cutkosky MR. DynaRing: a Patient Specific Mitral Annuloplasty Ring with Selective Stiffness Segments. *Journal of Medical Devices*, 2022. doi
6. **Kaiser AD**, Shad R, Hiesinger W, Marsden AL. A Design-Based Model of the Aortic Valve for Fluid-Structure Interaction. *Biomechanics and Modeling in Mechanobiology*, 2021. arXiv doi movies
5. Shad R, **Kaiser AD**, Kong S, Fong R, Quach N, Bowles C, Kasinpila P, Shudo Y, Teuteberg J, Woo YJ, Marsden AL, Hiesinger W. Patient Specific Computational Fluid Dynamics Reveal Localized Flow Patterns Predictive of Post-LVAD Aortic Incompetence. *Circulation: Heart Failure*, 2021. doi
4. Kasinpila P, Kong S, Fong R, Shad R, **Kaiser AD**, Marsden AL, Woo YJ, Hiesinger W. Use of Patient-Specific Computational Models for Optimization of Aortic Insufficiency after Implantation of Left Ventricular Assist Device. *The Journal of Thoracic and Cardiovascular Surgery*, 2020. doi
3. Bailey DH, Borwein JM, **Kaiser AD**. Automated Simplification of Large Symbolic Expressions. *Journal of Symbolic Computation*, 2014. doi

Conference proceedings (peer reviewed):

2. **Kaiser AD**, Williams S, Madduri K, Ibrahim K, Bailey DH, Demmel JW, Strohmaier E. A Principled Kernel Testbed for Hardware/Software Co-Design Research. *Proceedings of USENIX Workshop on Hot Topics in Parallelism*, 2010. pdf
1. **Kaiser AD**, Dolinar S, Cheng MK, Undetected Errors in Quasi-cyclic LDPC Codes Caused by Receiver Symbol Slips. *Proceedings of IEEE Global Conference on Communications*, 2009. doi

Submitted for review

6. Bornemann KM†, Choi PS, Huber JA, McMullen HL, Reed AK, Sharir A, Maskatia SA, Marsden AL, Ma MR, **Kaiser AD**. The Fluid Mechanics of Truncus Arteriosus. *Submitted.* 2026. arXiv
5. Reed AK, Martinez ES†, **Kaiser AD**, Sharir A, Bonin CC, Choi PS, Maskatia SA, Marsden AL, Ma MR. From Simulation to Suture Lines: Computational Fluid Dynamic Simulation-Guided Baffle Construction for Complex Biventricular Repair. *Submitted.* 2026. arXiv
4. Zhao H†, **Kaiser AD**, Kong F, Brown AL, Hu Z, Codoni D, Dave S, Marsden AL. Comparative analysis of resistive immersed surface and immersed boundary methods for aortic valve simulation. *Submitted.* 2026. arXiv

3. Hu Z, Rojas AFM, **Kaiser AD**, Zhao H†, Maskatia SA, Skylar-Scott MA, Humphrey JD, Hsia TY, Marsden AL. Patient-Specific Evaluation of a Confluence Pulsatile Conduit for Fontan Circulatory Support. *Submitted*. 2026.
2. Munshi SK, Huber JA, Ono Y, Reed AK, McMullen HL, Sharir A, Bonin CC, Boulos G, **Kaiser AD**, Ma MR. Ventricular Geometry and Valve Morphology Determine Atrioventricular Valve Regurgitation and Biomechanics in Single-Ventricle Physiology. *Submitted*. 2026.
1. Wang J, Brown AL, Zhao P, Zheng CZ, Chen JN, Zhou B, Liu J, Yokota T, **Kaiser AD**, Park SK, Marsden AL, Hsiai TK. A Continuum of Atrial Peristalsis Initiates the Bicuspid to Quadricuspid Valve Transition. *Submitted*. 2026. bioRxiv

Preprints & Technical Reports

2. Bao Y, **Kaiser AD**, Kaye J, Peskin CS. Gaussian-Like Immersed Boundary Kernels with Three Continuous Derivatives and Improved Translational Invariance. *Preprint*, 2017. arXiv
1. **Kaiser AD**, Williams S, Madduri K, Ibrahim K, Bailey DH, Demmel JW, Strohmaier E. TORCH – Computational Reference Kernels: A Testbed for Computer Science Research. *LBNL Technical Report*, 2010. pdf

Mentoring

Trainees mentored:

- **Karoline-Marie Bornemann, PhD**, Postdoctoral Scholar, Stanford University 2025 –
  Primary postdoctoral mentoring. Hired and fully funded under grants to ADK.
  Fluid-structure interaction modeling of neonates with rare congenital heart disease.
- **Camille Shantz, BA**, Medical Student, Stanford University 2026 –
  Primary mentoring of a Sarnoff Cardiovascular Research Foundation Fellow during a research year.
  Computational study of supravalvular aortic stenosis in children and its surgical treatment.
- **Elena Martinez, MS**, PhD Student, Stanford University 2025 –
  Co-mentoring with A. Marsden on mathematical modeling techniques and study design.
  Image-based modeling of complex congenital ventricular repair.
- **Han Zhao, PhD**, Postdoctoral Scholar, Stanford University 2025 – 2026
  Co-mentoring with A. Marsden. Fluid-structure interaction methods for valve simulations.
- **Sandra Kong**, Undergraduate, Stanford University 2018
  Mentoring of an undergraduate research assistant in a collaborator's laboratory.
  Modeling of vascular flows and medical devices for heart failure.

Resident and peer collaborators:

- **Perry Choi, MD**, Resident, Cardiothoracic Surgery, Stanford University 2023 –
  Collaborator and research support on funded internal postdoctoral fellowship.
- **Alexander Reed, MD**, Resident, Cardiothoracic Surgery, Stanford University 2025 –
  Collaborator and engineering advice on clinical translation of modeling complex ventricular repair.
- **Gabriel Weininger, MD**, Resident, Cardiothoracic Surgery, Stanford University 2025 –
  Collaborator on funded NIH postdoctoral fellowship on prosthetic valve design.
- **Jon Hochstein, MD**, Resident, Cardiothoracic Surgery, Stanford University 2025 –
  Collaborator on funded NIH postdoctoral fellowship on Fontan flow mechanics.
- **Hannah McMullen, MD**, Resident, Cardiothoracic Surgery, Stanford University 2026 –
  Collaborator and engineering advice on ex vivo experiments.
- **Rohan Shad, MD**, Postdoctoral Scholar, Cardiothoracic Surgery, Stanford University 2019 – 2022
  Collaborator on modeling congenital valvular disease and flows in heart failure.
- **Nicole Schiavone, MS**, PhD Student, Stanford University 2020 – 2021
  Collaborative study on experimental validation and flow analysis in immersed boundary methods.
- **Patpilai Kasinpila, MD**, Resident, Cardiothoracic Surgery, Stanford University 2018 – 2020
  Collaborator on modeling vascular flows and medical devices for heart failure.

Invited Seminars

15. "Designing Heart Valves from First Principles: Model Generation, Congenital Disease & Surgical Treatment," Biomedical Engineering Department Seminar, Tulane University, New Orleans LA, 2026
14. "Engineering Precision Medicine: Computational Modeling of Patient-Specific Surgical Outcomes," Heart Center Research Day, Stanford University, Stanford CA, 2026
13. "Designing Heart Valves from First Principles: Model Generation, Congenital Disease & Surgical Treatment," Mechanical Engineering and Applied Mechanics Department Seminar, University of Pennsylvania, Philadelphia PA, 2024
12. "Multimodal Optimization of Congenital Aortic Valve Repair," joint with Perry Choi, Biomechanics Seminar, Stanford University, Stanford CA, 2024
11. "Designing Heart Valves from First Principles: Model Generation, Congenital Disease & Surgical Treatment," HEART Lab Seminar, Division of Cardiovascular Medicine, Department of Medicine, Stanford University, Stanford CA, 2024
10. "Design-Based Models of Heart Valves for Fluid-Structure Interaction," Fluid Mechanics Seminar, Department of Mechanical Engineering, Stanford University, Stanford CA, 2022
9. "Design-Based Models of Heart Valves for Fluid-Structure Interaction," Computational Medicine Department Seminar, UCLA, Los Angeles CA, 2022
8. "Design-Based Models of Heart Valves for Fluid-Structure Interaction," Biomechanics and Medical Devices Seminar, Department of Mechanical Engineering, UC San Diego, San Diego CA (virtual), 2022
7. "Design-Based Models of Heart Valves for Fluid-Structure Interaction," Mathematics Department Seminar, UC Riverside, Riverside CA, 2022
6. "Design-Based Models of Heart Valves for Fluid-Structure Interaction," ReCoVor: Remote Colloquium on Vortex Flows, virtual, 2021
5. "Modeling the Mitral Valve," CME 300: First Year Seminar Series, Institute for Computational & Mathematical Engineering, Stanford University, Stanford CA, 2018
4. "Modeling the Mitral Valve," Computational Biology Colloquium, Courant Institute of Mathematical Sciences, New York University, New York NY, 2016
3. "Computational Experiments in Markov Chain Monte Carlo," Student Numerical Analysis and Scientific Computing Seminar, Courant Institute of Mathematical Sciences, New York University, New York NY, 2016
2. "Automated Simplification of Large Symbolic Expressions," Centre for Computer-Assisted Research Mathematics & its Applications Seminar, University of Newcastle, Newcastle NSW Australia, 2011
1. "A Testbed Based on the Motifs of Parallel Computing," Computational Research Division Seminar, Lawrence Berkeley National Laboratory, Berkeley CA, 2010

Software

- Primary author of open-source scientific software packages:
  - `heart_valves`: Model generation with elasticity-based design and fluid-structure interaction for aortic and mitral valves
  - `mc_stretch`: Affine-invariant, GPU-parallel MCMC sampler
  - `SimplifySum`: Automatic simplification of symbolic sums in *Mathematica*
- Collaborative development of scientific software libraries: IBAMR, SimVascular, ARPREC, QD, MatrixFFT

Teaching Experience

- Guest lecture, "Designing Heart Valves from First Principles" 2025
  Medicine 223: Cardiovascular and Pulmonary Sciences Seminar, Stanford University
- Guest lecture, "Designing Heart Valves from First Principles" 2025
  Mechanical and Aerospace Engineering 261: Cardiovascular Fluid Mechanics, University of California, San Diego
- Guest lecture, "Designing Heart Valves from First Principles" 2024
  Mechanical and Aerospace Engineering 261: Cardiovascular Fluid Mechanics, University of California, San Diego

- Guest lecture, "Newton's Method for Numerical Solution of Nonlinear Differential Equations" 2022
  Computational and Mathematical Engineering 209: Mathematical Modeling of Biological Systems, Stanford University
- Guest lecture, "Modeling Heart Valves" 2019
  Bioengineering 285: Computational Modeling in the Cardiovascular System, Stanford University
- Guest lecture, "A Model of the Mitral Valve & Patient-Specific Left-Ventricular Flow" 2019
  Medicine 289: Introduction to Bioengineering Research, Stanford University
- Teaching assistant, Mathematics GA 1410 2015
  Introduction to Mathematical Analysis I (graduate level), New York University
- Teaching assistant, Mathematics UA 325 2015
  Analysis (undergraduate level), New York University

Professional Service and Outreach

Service:

- **Journal peer review:** 2017 –
  Proceedings of the National Academy of Sciences (PNAS); Journal of Computational Physics; Physics of Fluids; Biomechanics and Modeling in Mechanobiology; Cardiovascular Engineering and Technology (CVET); International Journal for Numerical Methods in Biomedical Engineering; Journal of Biomechanical Engineering; Computers in Biology and Medicine; Journal of Mechanics in Medicine and Biology; Computer Graphics Forum; Scientific Reports.
- **Trainee committee:** International Conference on Functional Imaging and Modeling of the Heart 2021
- **Grant review:** Stanford Cardiovascular Institute Seed Grant Competition 2020
- **Session chair:** Fluid Dynamics I, Society for Industrial and Applied Mathematics Conference on the Life Sciences (cancelled) 2020
- **Organizer:** Graduate Student & Postdoc Seminar, Courant Institute 2015 – 2016

Leadership and outreach:

- **President:** Courant Student Organization 2015 – 2016
  - Represented concerns of the PhD student body anonymously to faculty.
  - Suggested improvements for recruitment and retention of women.
  - Recruitment subsequently increased: from one American woman among roughly 100 total PhD students in 2013 to four or more entering in 2017 alone.
- **PhD career panelist:** Bay Area Graduate Pathways to STEM conference 2018
- **Head coach:** Berkeley Ironworks Climbing and Fitness 2008 – 2009
  Head instructor of Berkeley Ironworks Teen Team, a non-competitive rock-climbing team for teenagers. Mentored over thirty teenagers, including some with physical disabilities such as cerebral palsy.
- **Presenter:** Oakland/East Bay Math Circle 2008

Conference Abstracts

9. Bornemann KM†, Choi PS, Huber JA, Sharir A, Maskatia SA, Ma MR, Marsden AL, **Kaiser AD**. Computational Modeling of a Quadricuspid Truncal Valve in Neonatal Hemodynamic Conditions. *American Physical Society Division of Fluid Dynamics Annual Meeting*, 2025. abstract
8. **Kaiser AD**, Choi PS, Sharir A, Marsden AL, Ma MR. Simulation-Guided Design of Leaflet Height in Bicuspidization of the Aortic Valve. *American Association for Thoracic Surgery 105th Annual Meeting*, 2025. abstract
7. Choi PS, Sharir A, Nilkant R, **Kaiser AD**, Ma MR. Biomechanical Assessment of No-cut Tricuspidization vs Bicuspidization Repairs in Quadricuspid Truncal Valve Model. *American Association for Thoracic Surgery 105th Annual Meeting*, 2025. abstract
6. Choi PS, Tarun S, Sharir A, Huber JA, Shibata M, Bornemann KM†, **Kaiser AD**, Marsden AL, Ma MR. Ex-vivo Assessment of Pulmonic Valve Mono-, Bi-, and Trileaflet Replacement using Autologous Pulmonary Arterial Wall. *KTCVS-STS Asia Symposium on Valvular Heart Disease, 26th Meeting*, 2025. program

5. Choi PS, Sharir A, Shibata M, **Kaiser AD**, Zhu Y, Marsden AL, Woo YJ, Ma MR, Kim JB. Effect of Graft Sizing in Valve-sparing Aortic Root Replacement for Bicuspid Aortic Valve: The Goldilocks Ratio. *American Association for Thoracic Surgery Aortic Symposium*, 2024. abstract
4. Haidar MA, Choi PS, **Kaiser AD**, Sharir A, Kapula N, Shibata M, Marsden AL, Ma MR. A Novel Approach for Aortic Valve Bicuspidization: Integrating Computational and Ex-vivo Simulation Concepts of Free-edge Leaflet Length in a modified Schäfer's procedure. *American Association for Thoracic Surgery 104th Annual Meeting*, 2024. abstract
3. **Kaiser AD**, Shad R, Schiavone N, Hiesinger W, Marsden AL. Fluid-Structure Interaction Simulations of Bicuspid Aortic Valve Disease. *Circulation 144 (Suppl_ 1)*, A13417, 2021. doi
2. **Kaiser AD**, Marsden AL. Modeling Patient-Specific Left-Ventricular Blood Flow and Mitral Valve Dynamics. *Proceedings of the 6th International Conference on Computational and Mathematical Biomedical Engineering (CMBE)*, extended conference abstract, 2019. pdf
1. Strohmaier E, Williams S, **Kaiser AD**, Madduri K, Ibrahim K, Bailey DH, Demmel JW. A Kernel Testbed for Parallel Architecture, Language and Performance Research. *Proceedings of the International Conference on Numerical Analysis and Applied Mathematics*, 2010. doi

Conference Presentations

28. "Combined Effect of Free Edge Length and Sinotubular Junction Diameter in Simulations of Bicuspidization Repair," Heart Center Research Day, poster, Stanford CA, 2026
27. "Simulation-Guided Design of Congenital Aortic Valve Repair," Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering (CSE25), Fort Worth TX, 2025
26. "Simulation-Guided Design of Leaflet Height in Bicuspidization of the Aortic Valve," American Association for Thoracic Surgery 105th Annual Meeting, Seattle WA, 2025
25. "Simulation-Guided Design of Leaflet Height in Bicuspidization of the Aortic Valve," Inaugural Stanford Cardiothoracic Surgery Annual Research Symposium, Stanford CA, 2025
24. "Simulation-Guided Design of Leaflet Height in Bicuspidization of the Aortic Valve," Bay Area Cardiovascular Research Symposium, Stanford CA, 2025
23. "Multimodal Optimization of Congenital Aortic Valve Repair," Stanford Maternal & Child Health Research Symposium, poster, Stanford CA, 2024
22. "Multimodal Optimization of Congenital Aortic Valve Repair," Stanford-Arizona-Morehouse-UAB Cardiovascular Research Symposium, poster, Stanford CA, 2024
21. "Multimodal Optimization of Congenital Aortic Valve Repair," Cardiac Imaging, Mechanics, and Modeling Symposium, Stanford CA, 2024
20. "Simulation-Based Design of Bicuspidization of the Aortic Valve," Stanford-Cornell-Penn-Duke Cardiovascular Research Symposium, poster, Stanford CA, 2023
19. "Topics in Fluid-Structure Interaction Involving Heart Valves: Repair of a Congenital Aortic Valve Defect," Cardiac Imaging, Mechanics, and Modeling Symposium, Stanford CA, 2023
18. "Controlled Comparison of Simulated Hemodynamics across Tricuspid and Bicuspid Aortic Valves," 18th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering (CMBBE), Paris, France, 2023
17. "Design-Based Models of Heart Valves for Fluid-Structure Interaction," Berkeley/Stanford Computational Mechanics Festival (CompFest), Stanford CA, 2022
16. "Controlled Comparison of Simulated Hemodynamics across Tricuspid and Bicuspid Aortic Valves," Stanford-Cornell Cardiovascular Research Symposium, poster, Stanford CA, 2022
15. "Controlled Comparison of Simulated Hemodynamics across Tricuspid and Bicuspid Aortic Valves," Cardiac Imaging, Mechanics, and Modeling Symposium, Stanford CA, 2022
14. "Fluid-Structure Interaction Simulations of Bicuspid Aortic Valve Disease," American Heart Association Scientific Sessions, virtual, 2021
13. "Design-Based Models of Heart Valves and Flow through Bicuspid Aortic Valves," 17th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering (CMBBE), poster, virtual, 2021
12. "Design-Based Models of Heart Valves and Bicuspid Aortic Valve Flows," Society for Mathematical Biology Annual Meeting, virtual, 2021

11. "A Design-Based Model of the Aortic Valve," Cardiac Imaging, Mechanics, and Modeling Symposium, Stanford University, virtual, 2020
10. "Simulating Patient-Specific Left-Ventricular Flow from Scan Data," Society for Industrial and Applied Mathematics Conference on the Life Sciences (cancelled), 2020
9. "Simulations of Patient-Specific Left-Ventricular Flow," Stanford-Penn Cardiovascular Symposium, poster, Stanford CA, 2019
8. "A Design-Based Model of the Mitral Valve & Simulations of Patient-Specific Left-Ventricular Flow," Mitral Day, Boston Children's Hospital at Harvard University, Boston MA, 2019
7. "Modeling Patient-Specific Left-Ventricular Blood Flow and Mitral Valve Dynamics," 6th International Conference on Computational and Mathematical Biomedical Engineering (CMBE), Tohoku University, Sendai, Japan, 2019
6. "A Design-Based Model of the Mitral Valve & Simulations of Patient-Specific Left-Ventricular Flow," Cardiac Imaging, Mechanics, and Modeling Symposium, Stanford CA, 2019
5. "Modeling the Mitral Valve," Stanford-Duke Cardiovascular Research Symposium, poster, Stanford CA, 2018
4. "Modeling the Mitral Valve," Stanford Bioengineering Department Retreat Symposium, poster, Santa Cruz CA, 2018
3. "SimCardio: Open-Source, Multi-Physics, Cardiac Modeling and Simulation," NSF Software Infrastructure for Sustained Innovation (SI2) Principal Investigator Meeting, poster, Washington DC, 2018
2. "Modeling the Mitral Valve," American Physical Society Division of Fluid Dynamics Annual Meeting, Portland OR, 2016
1. "Undetected Errors in Quasi-cyclic LDPC Codes Caused by Receiver Symbol Slips," IEEE Globecom, Honolulu HI, 2009

Additional Experience

UC Berkeley African Music Ensemble
Drummer 2007 – 2009
Recitation section leader 2007

- Lead drummer: ensemble of over one hundred people in rehearsal and performance.
- Volunteer teaching assistant: ran recitation sections and tutored students in drumming, dancing, and singing.

Drumming experience

- **African Music Ensemble** 2009 – 2011, 2017 –
  Performed traditional music of the Ewe people of Ghana. Led by C.K. Ladzekpo of UC Berkeley Music.
- **Cypress, Primes, Soft Signals, Scully**, rock bands 2011 – 2017
  Continued playing as a hobby.
- **Maus Haus**, rock band 2011
  Reviewed by *Rolling Stone*, *SF Weekly*.
- **Magic Bullets**, rock band 2010 – 2011
  Reviewed by *New York Magazine*, *Pitchfork*.
- **Tempo No Tempo**, rock band 2004 – 2010
  Reviewed favorably by *Rolling Stone*, *Pitchfork*. Voted "Best Student Band" at Berkeley.